15/09/21

Feature Reduction
- Feature Selection
  - Filter method
  - Wrapper method
    - Backward Selector
    - Forward Selector
- Feature Extraction

Forward Selector algo:

1) Start with an empty feature set
2) Choose each individual feature
3) Select features with maximum accuracy by using model trained on grow set and validated on validation set
4) Add one more feature and repeat 2, 3) and select the 2nd feature out of (m-1) combinations
5) Repeat the step 4) for next (p-2) feature

(OR)

Repeat until improvement of accuracy is less than the predefined threshold

Filtering method: (monovariate / univariate method)

1) Pearson's correlation coefficient (r)
2) F-Score
3) χ^2-method
4) SNR value

$$r_{(j)} = \frac{\sum_{i=1}^{n} (x_{ji} - \bar{x}_j)(\hat{y}_{ji} - \bar{y}_j)}{\sqrt{\sum (x_{ji} - \bar{x}_j)^2 \sum (y_{ji} - \bar{y}_j)^2}}$$

$r_{(j)}$ = correlation of jth input feature with target label

$r = [-1, 1]$

$r = 1 \Rightarrow$ highly correlated

$= -1 \Rightarrow$ hilghly correlated inversely

$= 0 \Rightarrow$ uncorrelated

$0 < r < 1 \Rightarrow$ correlated

$-1 < r < 0 \Rightarrow$ correlated inversely

Multivariat Selection (regression):

→ Consider all features

→ Compute vector W for any linear classifier (classification pt x is related with target as

$$w^T x + w_0 = y$$)

→ Discard small values of w as they will have less effect on the model

e.g: $w = [10 \quad 0.01 \quad -9]$

→ ranking = x_0, x_2, x_1

↳ we can drop based on limit or threshold

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Feature Extraction:

$$\begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{Bmatrix} \Longrightarrow \begin{Bmatrix} z_1 \\ z_2 \\ \vdots \\ z_p \end{Bmatrix} \qquad p < m$$

$$z_i = f[x_1, x_2 \cdots x_m]$$

1) Each feature is obtained as a function of original feature set.

Let W = projection matrix

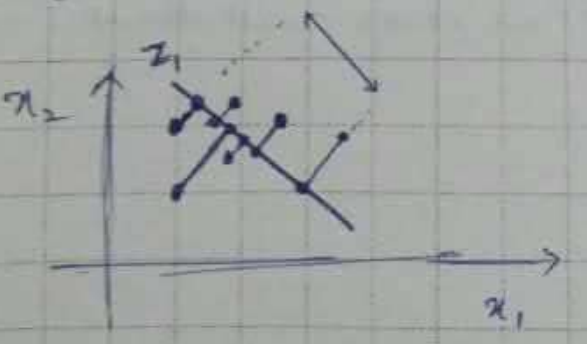
$\Rightarrow \bar{z} = W\bar{x}$

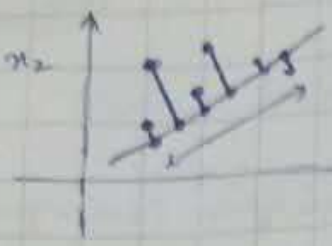
2) Projection of higher dim-feature set to lower dimension.

Essential properties ⇒ 1) New features should be uncorrelated

2) The variance of each feature should be maximised.

* To ensure maximum discriminance.





→ Also representing a 2D data to a single dimension

⇒ z_1' preferred over z_1, more spread

PCA (Principal Component Analysis) : $P = m$

↳ new features are linear combinations of original.

Algebraic definition of PCA ⇒

Given a set of 'n' observations of a vector of m variables (features)

$$\{x_1, x_2 \cdots x_m\} \in \mathbb{R}^m$$

define the first PC of the sample by linear transformation

$$z_1 = w_1^T x_j = \sum_{i=1}^{m} w_{1i} x_{ji} \quad ;$$

where $[w] = [w_{11}, w_{12} \cdots w_{1m}]$ is chosen such that

$var(z_1)$ is maximum

⇓

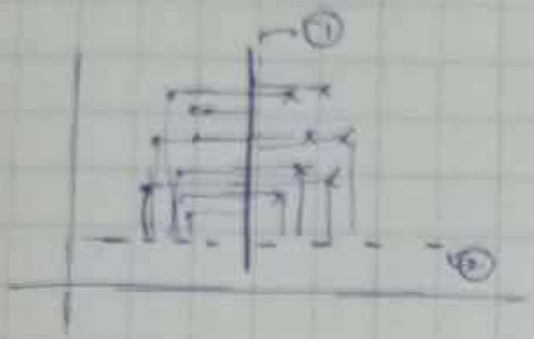
This also ensures reconstruction of dataset from the PC' projected dataset suffers minimum error

* First PC is chosen based on max variance.
2nd PC should be orthogonal to first PC

↓

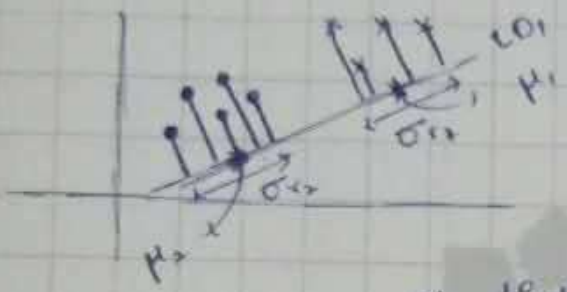
ensures zero correlation / uncorrelated

→ In PCA we never cared about output labels



according to PCA. ① is the first PC but mixing up the data

⇒ LDA (Linear discriminant analysis):



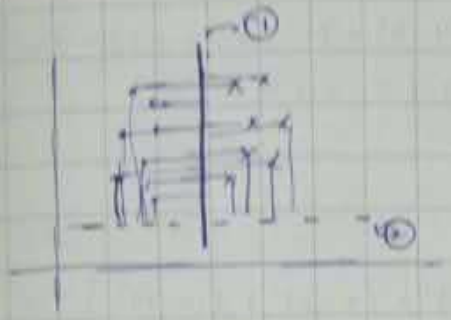
⇒ Inter cluster seperation is maximum

⇒ Intra cluster seperation is minimum

⇒ distance blw μ_1, μ_2 is max.

⇒ $\sigma_{r_1}, \sigma_{r_2}$ are minimum

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according to PCA. ① is the first PC that mixing up the data

LDA (linear discriminant analysis):



⇒ Inter cluster seperation is maximum

⇒ Intra cluster seperation is minimum

⇒ distance btw μ_1, μ_2 is max.

⇒ $\sigma_{x_1}, \sigma_{x_2}$ are minimum

ex:

	P_1	P_2	P_3	P_4
x_1	2	2	4	5
x_2	4	6	8	10

(2,4), (2,6), (4,8), (5,10)

Step 1 ⇒ form covariance matrix: (nxn matrix for n features)

$$S = \begin{bmatrix} cov(x_1,x_1) & cov(x_1,x_2) \\ cov(x_2,x_1) & cov(x_2,x_2) \end{bmatrix}$$

$$cov(x,y) = \frac{1}{N-1} \sum_{i=1}^{N} (x_{ix} - \bar{x})(y_{ix} - \bar{y}) \quad ; N = \text{total no. of datapoints}$$

$\bar{x}_1 = \frac{14}{4} = 3.5 \Rightarrow cov(x_1,x_1) = \frac{1}{3}\left((2-3.5)(2-3.5) + (3-3.5)^2 + (4-3.5)^2 + (5-3.5)^2\right)$

$\bar{x}_2 = ?$

$= \frac{5}{3}$

$$cov(x_1, x_2) = \frac{1}{3}\left((2-3.5)(4-7) + (3-3.5)(6-7) + (4-3.5)(8-7) + (5-3.5)(10-7)\right)$$
$$= cov(x_2, x_1)$$

$$= \frac{10}{3}$$

$$cov(x_2, x_2) = \frac{1}{3}\left((4-7)^2 + (6-7)^2 + (8-7)^2 + (10-7)^2\right)$$

$$= \frac{20}{3}$$

$$\Rightarrow S = \begin{bmatrix} \frac{5}{3} & \frac{10}{3} \\ \frac{10}{3} & \frac{20}{3} \end{bmatrix}$$

Step 2: Find eigen values of S =

$$q(\lambda) = |S - \lambda I| = \begin{bmatrix} \frac{5}{3} - \lambda & \frac{10}{3} \\ \frac{10}{3} & \frac{20}{3} - \lambda \end{bmatrix}$$

$$= \left(\frac{5}{3} - \lambda\right)\left(\frac{20}{3} - \lambda\right) - \frac{100}{9} = 0$$

$$= \frac{100}{9} - \frac{5}{3}\lambda - \frac{20}{3}\lambda + \lambda^2 = \frac{100}{9}$$

$$= \lambda^2 = \frac{25}{3}\lambda \Rightarrow \left(\lambda = 0 \;/\; \lambda = \frac{25}{3}\right)$$

largest eigen value is the largest, principal component first

Step 3: Find eigen vector

$\lambda = 0 \Rightarrow S - \lambda I = \begin{bmatrix} \frac{5}{3} & \frac{10}{3} \\ \frac{10}{3} & \frac{20}{3} \end{bmatrix}$

$\downarrow$

(λ_2)

$$\Rightarrow \bar{V}_2 = \begin{bmatrix} 20/3 \\ -10/3 \end{bmatrix}$$

$\lambda = \frac{25}{3} \Rightarrow S - \lambda I = \begin{bmatrix} -\frac{20}{3} & \frac{10}{3} \\ \frac{10}{3} & -\frac{5}{3} \end{bmatrix}$

$\downarrow$

λ_1

$\bar{v}_1 = \begin{bmatrix} -5/3 \\ -10/3 \end{bmatrix} \Rightarrow$ principal component axis

Step 4 : Normalization of eigen vector

$$\hat{v}_1 = \frac{\bar{v}_1}{\|\bar{v}_1\|} = \frac{[-5/3 \quad -10/3]^T}{\sqrt{\frac{25}{9} + \frac{100}{9}}} = \begin{bmatrix} -1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$$

Step 5 : Projection of points

$$v_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix} = [-0.45 \quad -0.89] \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix}$$

$(2,4) \rightarrow (-0.45)(2-3.5) - 0.89(4-7)$

$= 3.35$

$(3,6) \rightarrow 1.12$

$(4,8) \rightarrow -1.12$

$(5,10) \rightarrow -3.35$